

FIKIRU DEMLEW

Full-Stack Developer · Computer Vision & ML Engineer

+251974155283 · fikrudemlew@gmail.com · fikrudemlew.vercel.app · github.com/FikruDemlew · [LinkedIn](#) · Addis Ababa, Ethiopia

PROFILE

Graduated Mechatronics Engineer and full-stack developer with hands-on freelance experience delivering production-ready systems for international clients across Europe and Asia. Builds across the full stack — React and Vue frontends, Django and Go backends, and GraphQL with Hasura — while specialising in applied computer vision and ML: room geometry understanding with depth estimation, semantic segmentation using SAM2 and OneFormer, and AI-powered defect detection with YOLO and ONNX Runtime. Comfortable working in cross-functional remote teams, shipping complex pipelines end-to-end, and adapting across stacks to meet client requirements.

TECHNICAL SKILLS

Languages	JavaScript (ES6+), Python, Go, C++
Frontend	React, Next.js, Vue 3, HTML5, CSS3, Tailwind CSS
Backend / API	Django REST Framework, FastAPI, Node.js, Go (Gin), Hasura (GraphQL), REST, Swagger
Databases	PostgreSQL, SQLite, Redis, Firebase / Firestore
Real-time	WebSockets, Django Channels
Computer Vision	PyTorch, OpenCV, SAM2, OneFormer, Depth Anything V2, YOLO, ONNX Runtime
ML / AI	TensorFlow, Keras, CNN, MobileNetV2, Transfer Learning, Google Gemini AI
DevOps & Tools	Docker, Uvicorn, Postman, Git, GitHub, Netlify, Vercel
Payments	Third-party payment gateway integration
Soft Skills	Cross-cultural remote collaboration, Team leadership, Communication, Adaptability

WORK EXPERIENCE

Freelance Full-Stack & ML Developer · Remote — International Clients	2024 – Present
- Contracted by clients in Europe and Asia to build AI-powered systems and full-stack web applications, working within small cross-functional remote teams. - Delivered production-grade computer vision pipelines, REST APIs, and frontend interfaces across multiple independent engagements. - Managed project scoping, client communication, iterative delivery, and deployment independently and as part of small distributed teams.	

PROJECTS

AI Tile Visualizer — Room Geometry & CV Pipeline · Freelance · European Client · 5-member team	2024–2025
--	-----------

- Built a computer vision system that takes a single room photo as input, understands the room's geometry and wall layout, and realistically renders new tile designs onto the walls.
- Used SAM2 for precise wall and surface segmentation, handling complex occlusions from furniture, shadows, and overlapping objects.
- Applied OneFormer for semantic and panoptic segmentation to understand room structure and classify wall, floor, and ceiling regions accurately.
- Integrated Depth Anything V2 to estimate per-pixel depth from a single image, enabling perspective-correct tile rendering that respects the room's 3D geometry.
- Built an OpenCV-based pipeline for perspective correction and homography-based tile texture mapping onto segmented wall regions.
- Collaborated with a 4-member international team on model integration, rendering pipeline design, and iterative client feedback cycles.

Stack: *Python · PyTorch · SAM2 · OneFormer · Depth Anything V2 · OpenCV · Computer Vision · Geometry Processing*

AI Card Condition Grading System · [Freelance](#) · [Asian Client](#)

2025

- Developed an end-to-end AI system that analyses card images, detects surface and edge defects, evaluates centering accuracy, and generates an overall condition grade.
- Used YOLO for fast, accurate card and defect detection; exported models to ONNX Runtime for optimised, hardware-agnostic inference in production.
- Integrated Google Gemini AI to generate optional natural-language explanations of the grading result, giving end-users readable condition reports.
- Built a FastAPI backend with SQLAlchemy ORM, PostgreSQL/SQLite storage, and Redis caching for high-throughput image grading requests.
- Containerised the full stack with Docker and served via Uvicorn for reliable, scalable deployment.

Stack: *Python · FastAPI · PyTorch · OpenCV · YOLO · ONNX Runtime · Google Gemini AI · Redis · SQLAlchemy · PostgreSQL · SQLite · Docker · Uvicorn*

Freelance Marketplace Platform · [Full-Stack Project](#)

2025

- Built a full-stack platform connecting clients who post jobs with freelancers who browse, apply, and manage their work — role-based access enforced throughout.
- Clients manage jobs with category, budget, deadline, and status; freelancers submit and edit proposals with a streamlined application flow.
- Implemented real-time chat with Django Channels so clients and freelancers communicate directly within the platform without context-switching.
- Notification system alerts users about new applications, messages, and project updates in real time.
- [Live at freelance-marketplace-tau.vercel.app](https://freelance-marketplace-tau.vercel.app)

Stack: *React.js · Django REST Framework · Django Channels · PostgreSQL · Axios · Token Auth · Vercel*

YouTube Clone · [Personal Project](#)

2024

- Engineered a fully functional YouTube clone with video browsing, category filtering, and search via the YouTube Data API v3.
- Replicated YouTube's layout with a responsive, pixel-accurate React UI handling dynamic data fetching, loading states, and error boundaries.
- Architected clean component separation: sidebar, video cards, search bar, and video detail pages as isolated, reusable modules.
- [Live at youtubeclone1994.netlify.app](https://youtubeclone1994.netlify.app)

Stack: *React · Tailwind CSS · YouTube Data API v3 · Netlify*

Full-Stack Food Recipe Sharing Platform · [Personal Project](#)

2024–2025

- Built a full-stack recipe sharing site with Go backend, Hasura GraphQL over PostgreSQL, and a Vue 3 frontend.
- Containerised with Docker, API-documented with Swagger, and endpoint-tested with Postman.

Stack: *Vue 3 · Tailwind CSS · Go · Hasura (GraphQL) · PostgreSQL · Docker · Swagger · Postman*

E-Commerce Application with Payment Integration · [Personal Project](#)

2025

- Full-featured e-commerce app with product listings, cart, order flow, and live payment processing via a third-party gateway.

- PostgreSQL-backed REST API with JWT auth managing products, users, and orders.

Stack: *React · Node.js · PostgreSQL · REST API · Payment Gateway · JWT*

Delivery App — Frontend · [Google Developer Student Clubs \(GDSC\)](#) · **9-member team** 2023

- Built core UI components — order tracking, product catalogue, checkout — with mobile-first responsive design across all breakpoints.
- Integrated with a Go REST API, handling async data fetching, loading states, and error handling throughout the app.
- Contributed to team code reviews and maintained a shared component library standard across the codebase.

Stack: *React · Tailwind CSS · JavaScript · PostgreSQL · Go*

Tomato Quality Sorting AI — CNN + Transfer Learning · [ML Project](#) 2024

- Trained a MobileNetV2 CNN to classify tomatoes by quality grade; applied fine-tuning and augmentation on a custom dataset.

Stack: *Python · TensorFlow · Keras · MobileNetV2 · CNN · Transfer Learning*

Movie Watchlist Platform — Team Leader (5 members) · [Google Developer Student Clubs \(GDSC\)](#) 2022–2023

- Led a 5-member team to build a movie aggregation platform with trending data, search, and personalised watchlist backed by Firebase.

Stack: *HTML · CSS · JavaScript · React · Firebase*

EDUCATION

B.Sc. Mechatronics Engineering · [Addis Ababa Science and Technology University \(AASTU\)](#) 2021 – 2026

- Multidisciplinary programme: control systems, embedded systems, robotics, and software engineering.
- Active GDSC member — Frontend Developer & Team Leader.

Programming Fundamentals Nanodegree — Udacity

Verified Certificate · Python, problem-solving, and algorithmic thinking.

Machine Learning Course

Supervised learning, CNNs, and transfer learning — applied in tomato-sorting and tile-visualizer projects.